ResearchOS Evidence-Grounded Multi-Agent Architecture

Human-guided research coordination with traceable evidence, specialized agents, and claim-level verification



Researcher and Project Workspace

research question, sources, constraints



External Services

Model providers
Scholarly search
Code and data tools

models, search, and tools

RESEARCHOS AGENT RUNTIME

1 Evidence Foundation



Imported Sources



Project Library

Document Intelligence
parse · chunk · embed · provenance

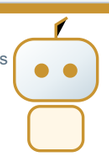
Hybrid Retrieval
dense + lexical + reranked evidence



Literature Agent
Discover and deduplicate studies
Synthesize evidence and prior work
Map the research landscape



Gap and Hypothesis Agent
Detect unresolved research questions
Generate falsifiable hypotheses
Track hypothesis lineage



Novelty and Feasibility
Compare contribution with prior art
Assess data, compute, risk, and ethics
Rank viable research directions



retrieval evidence packs



Critic and Writing Agent
Challenge claims and assumptions
Revise evidence-linked content
Compose academic sections

ResearchOS Orchestrator
compile · plan · route · monitor · replan



Experiment and Statistics
Design and execute analyses
Review methods and uncertainty
Capture reproducible provenance

claims, citations, and generated content

3 Verify and Deliver



Claim and Citation Validation
support · contradiction · completeness



Research Outputs
Draft and research chat
Verified references
Word and PDF export

verification failures return claims for retrieval, revision, or replanning

PERSISTENT RESEARCH PLATFORM



PostgreSQL
projects, runs, conversations



Vector Index
chunks, embeddings, provenance



Artifact Store
sources, analyses, drafts, exports



Security and Audit
tenant scope, approvals, usage

Solid navy: workflow · Solid teal: evidence flow · Coral dashed: verification feedback · Gray dashed: agent coordination and persistence